Chapter 2 — Force and Motion

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Scalars, vectors and rectangular components

Q.1 Differentiate scalar and vector quantities and explain how a vector is represented by rectangular components.

Scalar quantities have magnitude only, such as mass, time, temperature, energy and speed. Vector quantities have both magnitude and direction and obey the laws of vector addition, such as displacement, velocity, acceleration, force and momentum. A vector is drawn as an arrow whose length represents magnitude and whose arrowhead gives direction.

A vector A in a plane can be resolved into two perpendicular components: $A_x = A\cos\theta$ and $A_y = A\sin\theta$. The magnitude and direction are recovered from $A = \sqrt{(A_x^2 + A_y^2)}$ and $\tan\theta = A_y/A_x$. Components make vector addition easy because the x-components and y-components are added separately.

Feature	Scalar	Vector
Specification	Magnitude only	Magnitude and direction
Addition	Ordinary arithmetic	Triangle / parallelogram law or components
Examples	Mass, time, energy, speed	Displacement, velocity, force, momentum

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Important Conceptual Short Questions — Topic 1

1. Why is direction essential for a vector quantity?

The effect of a vector depends on its direction as well as its size. A force of 10 N to the right produces a different motion from a force of 10 N to the left, so direction must be specified.

2. How do rectangular components simplify vector addition?

Perpendicular components are independent. Adding all x-components and all y-components separately gives the components of the resultant, from which its magnitude and direction are found without drawing a scale diagram.

3. Why can a component of a vector not be greater than the vector itself?

A component is $A\cos\theta$ or $A\sin\theta$, and the sine or cosine of an angle cannot exceed 1. Therefore each rectangular component is at most equal to the magnitude of the vector.

Topic 2 — Product of two vectors

Q.2 Compare the dot product and cross product of two vectors with formulas, properties and examples.

The dot (scalar) product is $A \cdot B = AB \cos \theta$. It gives a scalar, is commutative, is maximum when the vectors are parallel and is zero when they are perpendicular. Work done by a constant force is an example: $W = F \cdot d = Fd \cos \theta$. The cross (vector) product is $A \times B = AB \sin \theta \, \hat{n}$, where $\hat{n}$ is a unit vector perpendicular to the plane of A and B given by the right-hand rule. It gives a vector, is anti-commutative, is maximum when the vectors are perpendicular and is zero when they are parallel. Torque $\tau = r \times F$ is an example.

Property	Dot product A·B	Cross product A×B
Result	Scalar AB cos θ	Vector AB sin θ n̂
Order	Commutative: A·B = B·A	Anti-commutative: A×B = −B×A
Parallel vectors	Maximum AB	Zero
Perpendicular vectors	Zero	Maximum AB
Example	Work W = F·d	Torque τ = r×F

Important Conceptual Short Questions — Topic 2

1. Why is the dot product of two perpendicular vectors zero?

The dot product contains cos θ, and cos 90° = 0. Physically, a force perpendicular to the displacement has no component along the displacement, so it does no work.

2. Why is the direction of the cross product perpendicular to both vectors?

The cross product represents an axis of rotation or a normal to the plane containing the two vectors. The right-hand rule fixes this direction uniquely, perpendicular to the plane of A and B.

3. How does the right-hand rule give the direction of A×B?

Point the fingers of the right hand along A and curl them toward B through the smaller angle. The extended thumb then points along A×B.

Topic 3 — Equations of motion

Q.3 Derive the three equations of uniformly accelerated motion by the graphical method.

For a velocity–time graph of uniform acceleration, the slope gives acceleration: $a = (v - u)/t$, so $v = u + at$. The displacement equals the area under the graph, which is a trapezium: $s = \frac{1}{2}(u + v)t$. Substituting $v = u + at$ gives $s = ut + \frac{1}{2}at^2$. Eliminating t between the first two equations gives $v^2 = u^2 + 2as$.

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These equations apply only when acceleration is constant and motion is along a straight line. The same equations can be used for vertical motion under gravity by taking a = g and choosing upward or downward as positive consistently.

Important Conceptual Short Questions — Topic 3

1. Why do the equations of motion apply only to uniform acceleration?

They are derived from a straight-line velocity–time graph whose slope is constant. If acceleration changes, the slope is not constant and the area and slope formulas used in the derivation no longer hold.

2. How does the area under a velocity–time graph give displacement?

Velocity is displacement per unit time. Multiplying velocity by a small time interval gives a small displacement, so summing all such strips, that is finding the area under the graph, gives the total displacement.

3. Why is the average velocity (u + v)/2 valid only for uniform acceleration?

For uniform acceleration the velocity increases linearly with time, so the mean of the initial and final velocities equals the time average. For non-uniform acceleration the graph is curved and this simple mean is not correct.

Topic 4 — Motion under gravity and projectile motion

Q.4 Analyze projectile motion and derive expressions for time of flight, maximum height and range.

A projectile has two independent motions. Horizontally there is no acceleration, so $v_x = v_0 \cos\theta$ remains constant. Vertically the acceleration is g downward, so $v_y = v_0 \sin\theta - gt$. The time of flight is obtained by setting the vertical displacement to zero: $T = 2v_0 \sin\theta/g$. The maximum height is reached when $v_y = 0$: $H = v_0^2 \sin^2\theta/(2g)$. The range is $R = v_x T = v_0^2 \sin 2\theta/g$.

The range is maximum when $\sin 2\theta = 1$, that is $\theta = 45°$. Two angles θ and $90° - \theta$ give the same range because $\sin 2\theta = \sin(180° - 2\theta)$. Air resistance reduces the range, height and time of flight and makes the path non-parabolic.

Important Conceptual Short Questions — Topic 4

1. Why does the horizontal velocity of a projectile remain constant?

In the absence of air resistance there is no horizontal force, so the horizontal acceleration is zero. By Newton’s first law the horizontal component of velocity remains unchanged.

2. Why do complementary angles give the same range?

The range depends on $\sin 2\theta$. Since $\sin 2\theta = \sin(180° - 2\theta)$, the angles θ and $90° - \theta$ give the same value of $\sin 2\theta$ and therefore the same range.

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3. How does air resistance affect projectile motion?

Air resistance opposes the motion, reducing both horizontal and vertical velocity components. The range, maximum height and time of flight become smaller, and the path is no longer a perfect parabola.

Topic 5 — Momentum, impulse and collisions

Q.5 State the principle of conservation of momentum and apply it to elastic and inelastic collisions.

The total momentum of an isolated system remains constant if no external force acts on it. This follows from Newton’s second and third laws: during a collision the forces between the bodies are equal and opposite, so the total change in momentum is zero. In an elastic collision both momentum and kinetic energy are conserved. In an inelastic collision momentum is conserved but kinetic energy is not; some energy becomes heat, sound or deformation. In a perfectly inelastic collision the bodies stick together and move with a common velocity.

Impulse is the product of force and the time for which it acts: $F\Delta t = \Delta p$. For a given change in momentum, increasing the time reduces the force. This is why a karate expert breaks bricks with a short sharp blow, while airbags and crumple zones increase the time of a car crash and reduce the force on passengers.

Important Conceptual Short Questions — Topic 5

1. Why does increasing the time of impact reduce the force?

Impulse equals the change in momentum, $F\Delta t = \Delta p$. For the same Δp, a larger Δt requires a smaller average force, so padding and airbags reduce injury.

2. Why is momentum conserved in a collision?

The colliding bodies exert equal and opposite forces on each other for the same time. Their impulses are equal and opposite, so the total momentum of the system does not change.

3. How does an elastic collision differ from an inelastic collision?

In an elastic collision kinetic energy is conserved as well as momentum. In an inelastic collision momentum is conserved but some kinetic energy is converted into other forms such as heat, sound and deformation.

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Topic 6 — Rocket propulsion

Q.6 Explain rocket propulsion and derive the acceleration of a rocket.

A rocket moves forward by ejecting hot gases backward at high speed. By conservation of momentum, the backward momentum of the ejected gases is balanced by the forward momentum gained by the rocket. If a mass m of gas is ejected with speed v relative to the rocket in a short time, and the mass of the rocket is M, the acceleration of the rocket is a = mv/M (where m is the mass ejected per unit time in the book's notation). The thrust is the rate of change of momentum of the gases.

A rocket does not need air to push against; it works in vacuum because the action–reaction pair is between the rocket and its own gases. As fuel burns, M decreases, so the same thrust produces a larger acceleration.

Important Conceptual Short Questions — Topic 6

1. Why can a rocket move in vacuum?

The rocket pushes its own gases backward, and the gases push the rocket forward. This action–reaction pair does not require any external medium, so the rocket works in empty space.

2. How does ejecting mass produce thrust?

Ejecting mass at high speed gives the gases backward momentum. Conservation of momentum requires the rocket to gain equal forward momentum, which appears as thrust.

3. Why does the acceleration of a rocket increase as fuel burns?

Thrust depends on the rate of ejection of gas, but the mass of the rocket decreases as fuel is used. Since a = F/M, a smaller M gives a larger acceleration for the same thrust.

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